NAME: ______________________

Instructor (circle one):
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MATH 2200, Calculus I, Fall 2018, Exam 4

Instructions:

- Show your work using the space provided on the exam. Use correct notation.
- Clearly mark each final answer by circling it.
- You may use a scientific calculator. But it is possible to complete the exam without one. Graphing calculators or other electronic devices, such as phones, are not allowed.
- Present your photo I.D. when turning in your exam.

Helpful Suggestions:

- You'll find two blank pages for scratchwork toward the end of the test.
- Briefly scan the exam and find problems that you feel most comfortable solving before attempting ones that seem harder. If you get stuck, move on; material in other problems may help get you unstuck.
- If a phrase or term is unclear, ask. To be fair, there is some material we can't explain while you're taking the exam, but you're welcome to ask.

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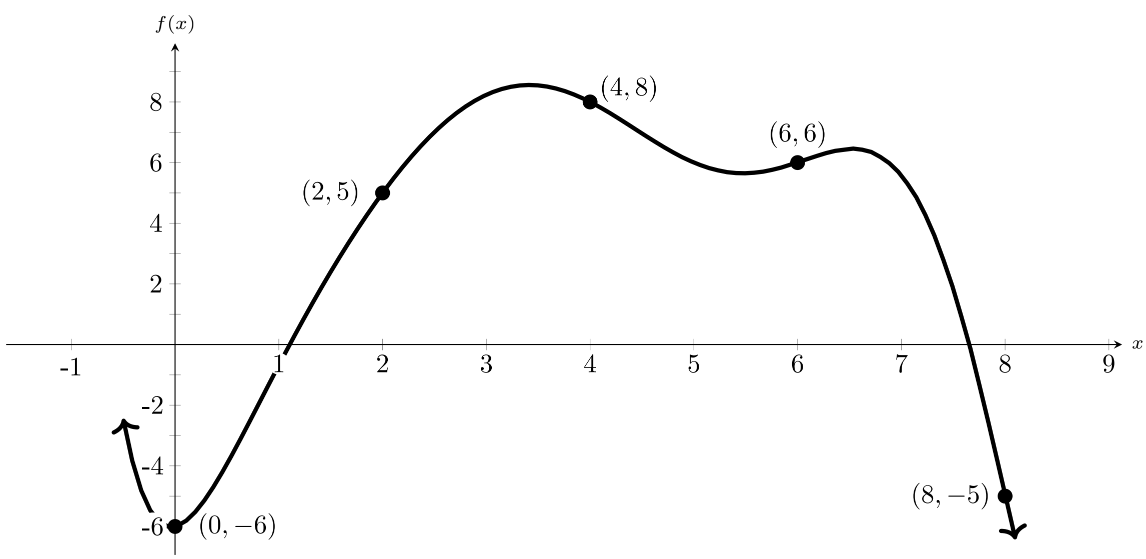
1. (15 points) Calculate the right Riemann sum on the interval $[0, 8]$ for the function graphed below. For your convenience, the set of points $\{0, 2, 4, 6, 8\}$ on the x-axis partitions the interval $[0, 8]$ into 4 subintervals of equal length, and the points $(x_i, f(x_i))$ corresponding to these values of x are shown on the graph of f given below.



2. (12 points) Let $f(x) = x^3 - 3x + 16$, defined for all real numbers x.

   (a) (3 points) Find all critical points of $f(x)$.

   (b) (3 points) Determine the interval(s) where f is increasing.

   (c) (3 points) Determine the interval(s) where f is decreasing.

   (d) (3 points) For each critical point found in part (a), determine whether it is a local maximum, a local minimum, or neither.

3. (18 points) Calculate the following integrals using the Fundamental Theorem of Calculus.

   (a) (6 points) $\displaystyle\int_{-1}^{1/2} \cos(\pi x)\, dx$

   (b) (6 points) $\displaystyle\int_{-1}^{2} e^{-2x}\, dx$

   (c) (6 points) $\displaystyle\int_{1}^{4} x^{5/2}\, dx$

4. (10 points) Calculate the following derivatives using the Fundamental Theorem of Calculus.

   (a) (5 points) $\displaystyle\frac{d}{dx}\int_{1}^{x} \frac{1}{2+\sin^3(t^2)}\, dt,\ x > 1.$

   (b) (5 points) $\displaystyle\frac{d}{dx}\int_{x}^{0} e^{-t^2}\, dt$

5. (9 points) Calculate the derivatives with respect to x of each of the following functions.

(a) (3 points) $f(x) = x^{\sin(x)}$ for $x > 0$.

(b) (3 points) $y(x)$, defined implicitly by the equation $x + y(x) = \exp(y(x))$ for $y \neq 0$..

(c) (3 points) $f(x) = \ln(x^3 + 3x^2 + 3x + 1)$, for $x \geqslant 0$.

6. (10 points) Find the average value of the function $f(x) = \sin(x)$ over the interval $[0, 2]$. (If you calculate a numerical value, remember that x is in radians.)

7. (12 points) Let f be defined on the real line by the formula

$$f(x) = \begin{cases} 0, \text{ if } x < -1, \\ |x| - 1, \text{ if } -1 \leqslant x \leqslant 1, \\ x^2 + 1, \text{ if } 1 < x. \end{cases}$$

(a) (4 points) At which points on the real line is f not continuous?

(b) (4 points) At which points on the real line is f not differentiable?

(c) (4 points) What is $\lim_{x \to 1^+} f(x)$?

8. (14 points) Using only symmetry (oddness/evenness) or geometry, evaluate the following integrals. Answers found using Riemann sums or the Fundamental Theorem of Calculus are not acceptable. **Justify your answer.**

   (a) (4 points) $\displaystyle\int_{-1}^{1} [x^3 + \sin(x)]\, dx.$

   Justification (3 points):

   (b) (4 points) $\displaystyle\int_{0}^{10} \frac{x}{5}\, dx.$

   Hint: You may find it useful to sketch a graph of the function being integrated.

   Justification (3 points):

9. (14 points) Evaluate the following limits using L'Hôpital's Rule where appropriate.

(a) $\lim_{x\to 0^+} \left(\dfrac{1}{x} - \dfrac{1}{e^x - 1} \right)$

(b) $\lim_{x\to 0^+} x^{\sqrt{x}}$.

SCRATCH WORK

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Differentiation formulas:

$f(u)$	$f'(u)$
$\tan u$	$\sec^2 u$
$\cot u$	$-\csc^2 u$
$\sec u$	$\sec u \tan u$
$\csc u$	$-\csc u \cot u$
$\ln f(u)$	$f'(u)/f(u)$
$\log_b u$	$1/(u \ln b)$
$\sin^{-1} u$	$1/\sqrt{1-u^2}$
$\tan^{-1} u$	$1/(1+u^2)$
$\sec^{-1} u$	$1/(\lvert u\rvert\sqrt{u^2-1})$
$\cos^{-1} u$	$-1/\sqrt{1-u^2}$
$\cot^{-1} u$	$-1/(1+u^2)$
$\csc^{-1} u$	$-1/(\lvert u\rvert\sqrt{u^2-1})$

Useful geometric formulas:

- Area of a circle having radius r: πr^2.
- Area of a triangle having base b and height h: $\frac{1}{2}bh$.
- Area of a trapezoid having bases b_1, b_2 and height h: $\frac{1}{2}(b_1 + b_2)h$.

For instructors' use only:

Problem	Value	Score
1	15	
2	12	
3	18	
4	10	
5	9	
6	10	
7	12	
8	14	
Total	100	